A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Redox equilibria

Every battery in your phone or car is a pair of redox half-reactions kept apart so that the electrons must travel through a wire. This topic shows how standard electrode potentials measure the pull of each half-reaction for electrons, how they predict which reactions can happen, and how redox titrations put oxidation and reduction to work in quantitative analysis.

Electrochemical cells

A **half-cell** is an element in two oxidation states in contact with each other, such as a strip of zinc dipping into a solution of zinc ions. Each half-cell sets up its own equilibrium, with the reduced form on the right by convention:

$$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s) \qquad Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$$

Zinc has a greater tendency than copper to lose electrons and form ions, so more electrons build up on the zinc strip. Join the two half-cells and there is a potential difference between them: the zinc is the **negative** electrode and the copper the **positive** one. For standard Zn and Cu half-cells it is 1.10 V.

- **Salt bridge**: completes the circuit by letting ions move between the two solutions, without mixing them. It is usually filter paper soaked in saturated potassium nitrate, because K^+ and NO_3^- do not react with the electrodes or solutions or form precipitates. (Potassium chloride would be unsuitable with silver or copper ions.) A metal wire cannot be used, because it would set up half-cells of its own.
- **High-resistance voltmeter**: draws almost no current, so the reactions do not proceed and the measured voltage is the maximum potential difference, the cell e.m.f.
- **When current flows** (a bulb instead of a voltmeter): the more positive electrode undergoes **reduction** ($Cu^{2+} + 2e^- \rightarrow Cu$), the more negative one undergoes **oxidation** ($Zn \rightarrow Zn^{2+} + 2e^-$), electrons flow through the wire from negative to positive, and the voltage gradually falls as the reactants are used up.

Cell diagrams

A cell is written in shorthand, with a single vertical line for a boundary between phases and a double line for the salt bridge. The oxidised forms are written next to the salt bridge, and by convention the positive electrode is on the right:

$$Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s) \quad E_{cell} = +1.10\ V$$

- **No metal in the half-reaction?** Use an inert **platinum** electrode, which conducts electrons but does not react. For $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ the half-cell is $\parallel Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt$, with a comma between species in the same phase and both ions at 1.00 mol dm^{-3}.
- **Gases** also need platinum: $\parallel Cl_2(g) \mid Cl^-(aq) \mid Pt$. H^+ and H_2O that only balance a half-equation are usually left out: $\parallel MnO_4^-(aq), Mn^{2+}(aq) \mid Pt$.

Standard electrode potentials

You cannot measure the potential of one half-cell on its own, only the difference between two. So every half-cell is measured against the same reference, the **standard hydrogen electrode** (SHE), whose potential is defined as exactly 0 V:

$$\text{Pt} \mid \text{H}_2(\text{g}) \mid \text{H}^+(\text{aq}) \qquad 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$$

- **The SHE**: hydrogen gas at 100 kPa bubbling over a platinum electrode coated in finely divided platinum (platinum black, which catalyses the equilibrium and adsorbs the gas), dipping into a solution with $[\text{H}^+] = 1.00$ mol dm^{-3} (usually 1.00 mol dm^{-3} HCl), at 298 K.
- **Standard electrode potential, $E^{\ominus}$**: the e.m.f. of a cell made from the half-cell and a standard hydrogen electrode, under standard conditions: 298 K, all solutions at 1.00 mol dm^{-3}, gases at 100 kPa, and no current flowing.
- **Why standard conditions?** Each half-cell is an equilibrium, so its potential changes with concentration and temperature. Adding more Zn^{2+} pushes $\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$ to the right, making its potential more positive.
- **Secondary standards**: the SHE is awkward to use, so ordinary measurements use electrodes calibrated against it, such as the silver/silver chloride electrode (+0.22 V) or the calomel electrode (+0.27 V).

Half-equation	$E^{\ominus}$ / V
$\text{Li}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Li}(\text{s})$	−3.03
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mg}(\text{s})$	−2.37
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$	−0.76
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	−0.44
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Ni}(\text{s})$	−0.25
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+0.34
$\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq})$	+0.54
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$	+0.80
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-(\text{aq})$	+1.09

$O_2(g) + 4H^+(aq) + 4e^- \rightleftharpoons 2H_2O(l)$	+1.23
$Cl_2(aq) + 2e^- \rightleftharpoons 2Cl^-(aq)$	+1.36
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$	+1.51
$F_2(g) + 2e^- \rightleftharpoons 2F^-(aq)$	+2.87

The more **negative** $E^\ominus$, the more readily the right-hand species loses electrons: the strongest **reducing agents** (Li, Mg) are at the top right. The more **positive** $E^\ominus$, the more readily the left-hand species gains electrons: the strongest **oxidising agents** (F_2, MnO_4^-) are at the bottom left.

Cell e.m.f. and feasibility

$$E^\ominus_{\text{cell}} = E^\ominus_{\text{right}} - E^\ominus_{\text{left}} = E^\ominus_{\text{reduced}} - E^\ominus_{\text{oxidised}}$$

In a working cell, the half-cell with the more positive $E^\ominus$ goes forwards (reduction) and the one with the more negative $E^\ominus$ goes backwards (oxidation). A reaction is **feasible** (thermodynamically possible) when $E^\ominus_{cell}$ for it is **positive**.

A nickel–silver cell

1. Ni^{2+}/Ni is −0.25 V and Ag^+/Ag is +0.80 V. Silver is more positive, so Ag^+ is reduced and nickel is oxidised.
2. $E^\ominus_{cell}$ = +0.80 − (−0.25) = **+1.05 V**. Cell diagram: Ni(s) | Ni^{2+}(aq) || Ag^+(aq) | Ag(s).
3. Combine the half-equations so the electrons cancel: $Ag^+ + e^- \rightarrow Ag$ is doubled, giving $Ni + 2Ag^+ \rightarrow Ni^{2+} + 2Ag$.

Never multiply $E^\ominus$ values when you double a half-equation: they measure a tendency per electron, not an amount.

Can iron(III) ions oxidise iodide? Bromide?

1. Fe^{3+}/Fe^{2+} is +0.77 V and I_2/I^- is +0.54 V. For Fe^{3+} to be reduced and I^- oxidised, $E^\ominus_{cell}$ = 0.77 − 0.54 = +0.23 V: **feasible**. $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$.

2 Br_2/Br^- is +1.09 V. For Fe^{3+} to oxidise Br^-, $E^{\ominus}_{cell}$ = 0.77 − 1.09 = −0.32 V: **not feasible**. The reverse is: bromine oxidises Fe^{2+} to Fe^{3+}.

Disproportionation

In **disproportionation** one species is both oxidised and reduced. Electrode potentials predict it: the species must appear on the left of one half-equation and the right of another with a less positive $E^{\ominus}$.

- **Copper(I)**: $Cu^+ + e^- \rightleftharpoons Cu$ (+0.52 V) and $Cu^{2+} + e^- \rightleftharpoons Cu^+$ (+0.15 V). $E^{\ominus}_{cell}$ = 0.52 − 0.15 = +0.37 V, so in aqueous solution $2Cu^+ \rightarrow Cu + Cu^{2+}$: copper goes from +1 to 0 and to +2. This is why copper(I) compounds such as Cu_2O give copper metal and a blue solution with dilute sulfuric acid.
- **Hydrogen peroxide**: $H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$ (+1.77 V) and $O_2 + 2H^+ + 2e^- \rightleftharpoons H_2O_2$ (+0.68 V). $E^{\ominus}_{cell}$ = +1.09 V, so $2H_2O_2 \rightarrow 2H_2O + O_2$, oxygen going from −1 to −2 and 0.

E_{cell}, entropy and the equilibrium constant

$E^{\ominus}_{cell}$ is directly proportional to the total entropy change, and to ln K, for the cell reaction:

$$E^{\ominus}_{\text{cell}} \propto \Delta S_{\text{total}} \propto \ln K$$

So a positive $E^{\ominus}_{cell}$ means a positive ΔS_{total} and K greater than 1, and the larger $E^{\ominus}_{cell}$ is, the further the reaction goes towards completion.

Limitations of predictions from $E^{\ominus}$

- **Kinetics**: a positive $E^{\ominus}_{cell}$ says a reaction *can* happen, not that it will be fast. If the activation energy is high it may be too slow to observe; copper does not react with water and oxygen at a noticeable rate, for example.
- **Non-standard conditions**: real concentrations, temperatures and pressures shift the electrode potentials (Le Chatelier applied to each half-equation). Increasing the concentration of a reactant in the cell reaction, or decreasing a product, increases E_{cell}; most cell reactions are exothermic, so raising the temperature usually lowers it. When $E^{\ominus}_{cell}$ is small (say under about 0.3 V), the change can even reverse the prediction.
- **Current flowing**: once a cell is used, the reactants are consumed and E_{cell} falls to zero when the cell reaction reaches equilibrium.

Fuel cells

A **fuel cell** converts the energy of a fuel reacting with oxygen directly into electricity. Unlike an ordinary battery, the reactants are fed in continuously, so the voltage stays constant instead of falling as the reactants are used up.

Hydrogen–oxygen fuel cell	Alkaline electrolyte (KOH)	Acid electrolyte
negative electrode (H_2 fed in)	$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$ ($E^\ominus$ −0.83 V)	$H_2 \rightarrow 2H^+ + 2e^-$ ($E^\ominus$ 0.00 V)
positive electrode (O_2 fed in)	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ ($E^\ominus$ +0.40 V)	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ ($E^\ominus$ +1.23 V)
overall	$2H_2 + O_2 \rightarrow 2H_2O$, $E^\ominus_{cell}$ = +1.23 V either way	

- **Advantages** over petrol or diesel engines: the only product is water, so no CO_2 or other pollutants are released where the fuel is used, and they convert energy more efficiently.
- **Limitations**: hydrogen is a flammable gas that is hard to store and transport (as a liquid under pressure, or adsorbed onto or absorbed into solids); fuel cells are expensive and have limited lifetimes; and most hydrogen is made from fossil fuels or by electrolysis, so it is only "green" if that electricity comes from renewable sources.
- **Ethanol fuel cells** avoid the storage problem: ethanol is a liquid that can be made from crops. At the negative electrode $C_2H_5OH + 3H_2O \rightarrow 2CO_2 + 12H^+ + 12e^-$; overall $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. Methanol can be used in the same way.

Redox titrations with potassium manganate(VII)

Acidified potassium manganate(VII) is a powerful oxidising agent that acts as its own indicator: purple MnO_4^- is reduced to almost colourless Mn^{2+}. Put it in the burette; the end point is the first permanent pale pink colour, showing a tiny excess of MnO_4^-. Read the burette from the top of the meniscus, because the dark purple solution hides the bottom.

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

COMMON MISTAKE

Use dilute sulfuric acid, in excess. Too little acid, or a weak acid such as ethanoic acid, cannot supply the $8H^+$ needed, and brown MnO_2 forms instead, hiding the end point. Hydrochloric acid is unsuitable because MnO_4^- oxidises Cl^- to chlorine ($E^\ominus_{cell}$ = 1.51 − 1.36 = +0.15 V), using up extra manganate(VII) and making a toxic gas. Nitric acid is unsuitable because it is an oxidising agent itself and would oxidise some of the Fe^{2+} (NO_3^-/HNO_2 +0.94 V > Fe^{3+}/Fe^{2+} +0.77 V), so too little manganate(VII) would be used.

Iron in iron tablets

1. 2.80 g of iron tablets is dissolved in dilute sulfuric acid and made up to 250 cm^3. 25.0 cm^3 portions need 24.60 cm^3 of 0.00500 mol dm^{-3} $KMnO_4$. Find the percentage of iron by mass.
2. Moles of MnO_4^- = 0.02460 × 0.00500 = 1.23×10^{-4} mol.
3. The equation has 5 Fe^{2+} per MnO_4^-: moles of Fe^{2+} in 25.0 cm^3 = $5 \times 1.23 \times 10^{-4} = 6.15 \times 10^{-4}$ mol.
4. In the whole 250 cm^3: 6.15×10^{-3} mol, so the mass of iron = $6.15 \times 10^{-3} \times 55.8$ = 0.343 g.
5. Percentage of iron = 0.343 ÷ 2.80 × 100 = **12.3%**.

- **Hydrogen peroxide**: $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$ (ratio 2 : 5).
- **Ethanedioate**: $2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$ (ratio 2 : 5). This reaction between two negative ions starts slowly, so the flask is warmed to about 60 °C; the Mn^{2+} formed then catalyses it.
- **Iron(II) ethanedioate**, FeC_2O_4: both ions are oxidised. One FeC_2O_4 gives up 1 + 2 = 3 electrons and each MnO_4^- takes 5, so the ratio is $3MnO_4^-$: $5FeC_2O_4$.
- For any other redox titration, write the two half-equations, balance the electrons, and use the overall ratio.

Iodine–thiosulfate titrations

Iodine reacts quickly and completely with sodium thiosulfate, which is used to measure how much iodine a reaction has made:

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

This lets you find the amount of almost any **oxidising agent**: add it to excess potassium iodide, so it oxidises I^- to I_2, then titrate that iodine with standard thiosulfate.

- **Starch indicator**: the iodine solution starts yellow-brown and fades to pale yellow as thiosulfate is added. Only then add a little starch, which forms an intense blue-black complex with the remaining iodine. The end point is when the blue-black colour disappears, leaving a colourless solution.
- **Why add starch late?** With a lot of iodine present the starch–iodine complex is insoluble and does not release its iodine readily, so the end point would be inaccurate.

Oxidising agent	Reaction with excess I^-	Overall ratio
copper(II) ions	$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$ (white CuI precipitate)	1 Cu^{2+} : 1 $S_2O_3^{2-}$

iodate(V) ions, acidified	$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$	1 IO_3^- : 6 $S_2O_3^{2-}$
chlorate(I) in bleach, acidified	$ClO^- + 2I^- + 2H^+ \rightarrow I_2 + Cl^- + H_2O$	1 ClO^- : 2 $S_2O_3^{2-}$

Copper in brass

1. 1.50 g of brass is dissolved in nitric acid, neutralised and made up to 250 cm^3. A 25.0 cm^3 portion is added to excess KI, and the iodine released needs 15.70 cm^3 of 0.100 mol dm^{-3} sodium thiosulfate. Find the percentage of copper.
2. Moles of $S_2O_3^{2-}$ = 0.01570 × 0.100 = 1.57×10^{-3} mol.
3. $2Cu^{2+}$ make 1 I_2, which reacts with 2 $S_2O_3^{2-}$, so Cu^{2+} : $S_2O_3^{2-}$ = 1 : 1. Moles of Cu^{2+} in 25.0 cm^3 = 1.57×10^{-3} mol.
4. In 250 cm^3: 1.57×10^{-2} mol, so mass of copper = $1.57 \times 10^{-2} \times 63.5$ = 0.997 g.
5. Percentage of copper = 0.997 ÷ 1.50 × 100 = **66.5%**.

Uncertainty in titrations

- **A reading** is one judgement on a scale; **a measurement** is the difference between two readings. A titre is a measurement, so a burette with ±0.05 cm^3 per reading gives ±0.10 cm^3 on each titre.
- $\text{percentage uncertainty} = \dfrac{\text{uncertainty}}{\text{quantity measured}} \times 100$. For a 24.60 cm^3 titre: 0.10 ÷ 24.60 × 100 = 0.41%. Add the percentage uncertainties of every piece of apparatus to estimate the total.
- **Reducing it**: measure a larger quantity (a bigger titre, by using a more concentrated solution in the flask or a more dilute one in the burette), or use apparatus with a smaller uncertainty, such as a pipette instead of a measuring cylinder.
- **Comparing with an accepted value**: percentage difference = (difference ÷ accepted value) × 100. If it is larger than the total apparatus uncertainty, there must be other errors in the method; if it is smaller, the apparatus uncertainty explains the gap.

Quick checks

QUICK CHECK

What is $E^{\ominus}_{cell}$ for Mg(s) | Mg^{2+}(aq) || Cu^{2+}(aq) | Cu(s)?

- −2.71 V
- −2.03 V
- +2.71 V
- +1.71 V

$E^{\ominus}_{cell} = E_{right} - E_{left} = +0.34 - (-2.37) = +2.71$ V. −2.03 V comes from adding the two values instead of subtracting; −2.71 V from subtracting the wrong way round.

QUICK CHECK

Why is the salt bridge usually soaked in potassium nitrate rather than potassium chloride?

- KNO_3 conducts electrons better
- Chloride ions could react with some solutions, for example precipitating Ag^+ as AgCl
- KCl would react with the voltmeter
- Nitrate ions take part in the cell reaction

The salt bridge carries ions, not electrons, and its ions must not react with either half-cell.

QUICK CHECK

Which statement about the standard hydrogen electrode is correct?

- it uses hydrogen at 1 atm and 1.00 mol dm^{-3} NaOH
- its potential is measured as 0 V against a calomel electrode
- it uses H_2 at 100 kPa, 1.00 mol dm^{-3} H^+ and a platinum electrode at 298 K
- the platinum is there to react with the hydrogen

Its potential is defined as 0 V, not measured. The platinum black is an inert, conducting surface that catalyses the H_2/H^+ equilibrium.

QUICK CHECK

Using the table, which species could oxidise Br^- to Br_2 under standard conditions?

- Fe^{3+}
- I_2
- Cu^{2+}
- Cl_2

The oxidising agent must have a more positive $E^{\ominus}$ than Br_2/Br^- (+1.09 V). Cl_2/Cl^- is +1.36 V, giving $E^{\ominus}_{cell}$ = +0.27 V; the others give negative values.

QUICK CHECK

$E^{\ominus}_{cell}$ for a reaction is +0.45 V, but no reaction is seen. What is the best explanation?

- the activation energy is too high, so the reaction is too slow
- the reaction is not feasible
- $E^{\ominus}_{cell}$ should be negative for a reaction to occur
- K is less than 1

A positive $E^{\ominus}_{cell}$ means the reaction is feasible and K > 1, but it says nothing about the rate.

QUICK CHECK

Why is hydrochloric acid not used to acidify potassium manganate(VII) in titrations?

- it is too weak to supply enough H^+
- it reduces Fe^{3+} back to Fe^{2+}
- MnO_4^- oxidises Cl^- to Cl_2, so more manganate(VII) is used
- it forms a precipitate with Mn^{2+}

$E^{\ominus}(MnO_4^-/Mn^{2+})$ = +1.51 V is more positive than $E^{\ominus}(Cl_2/Cl^-)$ = +1.36 V, so the titre would be too large and toxic chlorine would form.

QUICK CHECK

0.100 g of potassium iodate(V), KIO_3 (M_r 214), is added to excess acidified KI. What volume of 0.100 mol dm^{-3} thiosulfate reacts with the iodine formed?

- 4.67 cm^3
- 28.0 cm^3
- 14.0 cm^3
- 9.35 cm^3

Moles of IO_3^- = 0.100 ÷ 214 = 4.67 × 10^{-4}. Each makes $3I_2$, which need 6 $S_2O_3^{2-}$: 2.80 × 10^{-3} mol, which is 28.0 cm^3. 14.0 cm^3 forgets that each I_2 needs two thiosulfate ions.

QUICK CHECK

In an iodine–thiosulfate titration, when should the starch be added?

- at the start, so the colour change is easier to see
- near the end point, when the iodine colour has faded to pale yellow
- after the end point, to confirm it
- it is not needed because thiosulfate is coloured

Added early, the starch–iodine complex forms in large amounts and does not release its iodine readily, giving an inaccurate end point.

A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Transition metals and their chemistry

Copper sulfate is blue, potassium manganate(VII) purple, and iron catalyses the Haber process. These properties all come from one feature of the transition metals: an incomplete set of d electrons. This topic explains complex ions, their shapes and colours, the many oxidation states of vanadium and chromium, the reactions of metal aqua ions, and how transition metals act as catalysts.

What makes a transition metal

A **transition metal** is a d-block element that forms one or more stable ions with an **incomplete d sub-shell**. Across the period from scandium to zinc, electrons are added to the 3d sub-shell, but when the atoms form ions they lose their **4s electrons first**.

Element	Atom	Common ion	Ion configuration
Ti	[Ar] $3d^2\ 4s^2$	Ti^{3+}	[Ar] $3d^1$
Cr	[Ar] $3d^5\ 4s^1$	Cr^{3+}	[Ar] $3d^3$
Fe	[Ar] $3d^6\ 4s^2$	Fe^{2+}, Fe^{3+}	[Ar] $3d^6$, [Ar] $3d^5$
Cu	[Ar] $3d^{10}\ 4s^1$	Cu^{2+}	[Ar] $3d^9$

- **Chromium and copper** have only one 4s electron: a half-filled ($3d^5$) or full ($3d^{10}$) d sub-shell is especially stable.
- **Scandium is not a transition metal**: its only ion, Sc^{3+}, is [Ar] with an *empty* 3d sub-shell.
- **Zinc is not a transition metal**: its only ion, Zn^{2+}, is [Ar] $3d^{10}$, a *full* d sub-shell.
- Because the 4s and 3d electrons are close in energy, successive ionisation energies rise gradually until all of them are removed, so several oxidation states are reachable.
- **Characteristic properties**: they form **complex ions**, form **coloured** compounds, show **variable oxidation states**, and act as **catalysts**.

Complex ions and ligands

- **A complex ion** is a central metal ion surrounded by ligands bonded to it by dative (coordinate) bonds.
- **A ligand** is a molecule or ion with a lone pair of electrons that it donates to the metal ion to form a dative covalent bond, in which both shared electrons come from the same atom.
- **Coordination number**: the number of dative bonds to the central metal ion (not the number of ligands).

- **Monodentate** ligands form one dative bond: H_2O, NH_3, Cl^-, OH^-, CN^-.
- **Bidentate** ligands have two atoms that each donate a lone pair: ethane-1,2-diamine, $H_2NCH_2CH_2NH_2$ ("en"), and the ethanedioate ion, $C_2O_4^{2-}$. Three of them give a coordination number of 6, as in $[Cr(en)_3]^{3+}$ or $[Cr(C_2O_4)_3]^{3-}$.
- **Multidentate** ligands form several bonds: $EDTA^{4-}$ donates from two nitrogen atoms and four oxygen atoms, wrapping around a metal ion to form six bonds in a 1 : 1 complex. Haem is a multidentate ligand holding iron(II) in haemoglobin.
- **Charge of a complex** = charge of the metal ion + charges of the ligands: Cu^{2+} with four Cl^- gives $[CuCl_4]^{2-}$.

Shapes of complex ions

Shape	Coordination number	Bond angle	Examples
octahedral	6	90°	$[Cu(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$; most complexes with small ligands
tetrahedral	4	109.5°	$[CuCl_4]^{2-}$, $[CoCl_4]^{2-}$: chloride ions are larger, so fewer fit around the metal
square planar	4	90°	cisplatin, $[Pt(NH_3)_2Cl_2]$
linear	2	180°	$[Ag(NH_3)_2]^+$ (in Tollens' reagent), $[Ag(S_2O_3)_2]^{3-}$, $[Ag(CN)_2]^-$

Cisplatin and haemoglobin

Cisplatin is a square planar platinum(II) complex with the two ammonia ligands next to each other (cis) and the two chloride ligands next to each other. Its isomer with them opposite each other, transplatin, has no anticancer action. In the body a chloride ligand is replaced by water, $[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2Cl(H_2O)]^+ + Cl^-$, and the platinum then forms dative bonds to nitrogen atoms on neighbouring bases in DNA, kinking the strand so the cancer cell cannot copy its DNA and divide. It also binds to the DNA of healthy cells, which causes side effects such as hair loss.

Haemoglobin carries oxygen as a ligand bonded to Fe^{2+} in haem. Carbon monoxide is toxic because it forms a much stronger dative bond to the iron, replacing oxygen in a ligand exchange that is hard to reverse, so less oxygen can be carried.

Why transition metal compounds are coloured

In an isolated ion the five 3d orbitals have the same energy. When ligands bond to the ion, their lone pairs repel the d electrons, and in an octahedral complex the d orbitals **split into two groups** at different energies, separated by

ΔE. A d electron can absorb a photon of visible light whose energy exactly matches ΔE and jump to the upper group:

$$\Delta E = hf$$

The light that is absorbed is removed from white light, and the colour we see is the **complementary colour** of the light that is transmitted or reflected. Hydrated copper(II) ions absorb in the orange-red region, so they look blue.

- **What changes the colour**: anything that changes ΔE: a different **ligand**, a different **coordination number** (and so shape), or a different **oxidation state** of the metal.
- **Colourless ions**: Sc^{3+} ($3d^0$) has no d electrons to promote, and Zn^{2+} and Cu^{+} ($3d^{10}$) have no space in the upper d orbitals, so they absorb no visible light.

Measuring concentration by colorimetry

The amount of light a coloured solution absorbs is proportional to the concentration of the coloured species. A colorimeter (spectrophotometer) uses a filter of the complementary colour, the colour the solution absorbs most strongly. If a complex is only pale, first add a ligand that forms a more intensely coloured complex. Make up solutions of known concentration, measure their absorbance, and plot a **calibration curve**; then measure the unknown and read its concentration from the graph.

Variable oxidation states

The 4s and 3d electrons are similar in energy, so transition metals can lose different numbers of them. Some general trends: high oxidation states such as manganese(VII) in MnO_4^- are oxidising agents; low ones such as V^{2+} and Fe^{2+} are reducing agents; and the +2 state becomes more stable relative to +3 across the period.

Vanadium

Oxidation state	Ion	Colour in solution
+5	VO_2^+ (from ammonium vanadate(V), NH_4VO_3, in acid)	yellow
+4	VO^{2+}	blue
+3	V^{3+}	green
+2	V^{2+}	violet

Adding zinc to acidified vanadate(V) reduces it step by step, through yellow, then green (a mixture of yellow and blue), blue, green and finally violet. Electrode potentials explain how far a reducing agent can go:

- $VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$, +1.00 V; $VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$, +0.34 V; $V^{3+} + e^- \rightleftharpoons V^{2+}$, −0.26 V.
- **Zinc** (Zn^{2+}/Zn −0.76 V) is more negative than all three, so it reduces vanadium all the way to V^{2+}.
- **Tin** (Sn^{2+}/Sn −0.14 V) is more negative than the first two but not the third, so it stops at V^{3+}.

Chromium

- **Reducing dichromate(VI)**: zinc and hydrochloric acid reduce orange $Cr_2O_7^{2-}$ to green Cr^{3+}, then to blue Cr^{2+}: $Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$, then $2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$. Air must be kept out (for example with a stream of hydrogen), because oxygen oxidises Cr^{2+} straight back to Cr^{3+}.
- **Iron(II) is a weaker reducing agent**: $E^{\ominus}(Fe^{3+}/Fe^{2+})$ = +0.77 V lies between $E^{\ominus}(Cr_2O_7^{2-}/Cr^{3+})$ = +1.33 V and $E^{\ominus}(Cr^{3+}/Cr^{2+})$ = −0.41 V, so Fe^{2+} reduces dichromate only as far as Cr^{3+}. $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$ is used in titrations (with an indicator, as the colour change is not sharp enough on its own).
- **Oxidising chromium(III)** is easier in alkaline solution, because removing electrons from a negative ion is easier. With excess NaOH, Cr^{3+} forms green $[Cr(OH)_6]^{3-}$, which hydrogen peroxide oxidises to yellow chromate(VI): $2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$.
- **Chromate and dichromate** are linked by an acid–base equilibrium, not a redox one (chromium is +6 in both): $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$. Adding acid turns yellow chromate orange; adding alkali removes H^+ and turns it back.

Writing a half-equation in alkaline conditions

1. Start with the change: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-}$.
2. Balance oxygen with water and hydrogen with H^+, as if in acid: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+$.
3. Balance charge with electrons: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+ + 3e^-$.
4. Add OH^- to both sides to turn each H^+ into water: $[Cr(OH)_6]^{3-} + 2OH^- \rightarrow CrO_4^{2-} + 4H_2O + 3e^-$.

Reactions of metal aqua ions

A **Lewis acid** accepts an electron pair; a **Lewis base** donates one. In a complex the ligands are Lewis bases and the metal ion is a Lewis acid. Dissolved transition metal salts form hexaaqua ions, $[M(H_2O)_6]^{2+}$ or $[M(H_2O)_6]^{3+}$.

Acidity of aqua ions

The metal ion pulls electron density from the oxygen of each water ligand, weakening its O–H bonds, so a proton can be lost to the solvent:

$$[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$$

3+ ions are smaller and more highly charged than 2+ ions, so they have a greater **charge density** and polarise the water ligands more strongly. Solutions of M^{3+} ions are therefore noticeably acidic (around pH 2 to 3), while M^{2+} solutions are only very weakly acidic. This hydrolysis is why iron(III) solutions look yellow-brown rather than pale violet.

Deprotonation: hydroxide precipitates

Hydroxide ions and ammonia act as **Brønsted–Lowry bases**, removing protons from water ligands one at a time until a neutral complex forms, which precipitates:

- $[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(OH)_2(H_2O)_4] + 2H_2O$ (blue precipitate)
- $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(OH)_2(H_2O)_4] + 2NH_4^+$ (the same precipitate with a little ammonia)
- $[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(OH)_3(H_2O)_3] + 3H_2O$ (orange-brown precipitate)

Amphoteric hydroxides

The hydroxides of chromium(III) and zinc dissolve in excess sodium hydroxide, because further deprotonation gives a soluble negative ion; they also dissolve in acid. A hydroxide that reacts with both acids and alkalis is **amphoteric**:

- $[Cr(OH)_3(H_2O)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$ (dark green solution)
- $[Zn(OH)_2(H_2O)_4] + 2OH^- \rightarrow [Zn(OH)_4]^{2-} + 4H_2O$ (colourless solution)
- With acid: $[Cr(OH)_3(H_2O)_3] + 3H^+ \rightarrow [Cr(H_2O)_6]^{3+}$.

Ligand exchange

- **With excess ammonia**, NH_3 replaces water (and hydroxide) ligands on copper, cobalt, nickel, chromium(III) and zinc, dissolving the precipitate. NH_3 and H_2O are similar in size and both uncharged, so the coordination number stays 6 (exchange may be incomplete, as with copper): $[Cu(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2H_2O + 2OH^-$ (deep blue).
- **With concentrated HCl** (a high concentration of Cl^-), chloride replaces water. Cl^- is larger than H_2O, so only four fit and the coordination number falls from 6 to 4, giving tetrahedral complexes: $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$ (yellow-green); $[Co(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CoCl_4]^{2-} + 6H_2O$ (pink to blue). Adding water reverses these.

Stability of complexes and the chelate effect

Replacing monodentate ligands with bidentate or multidentate ones gives a much more stable complex. This **chelate effect** is explained by entropy:

- $[Cu(H_2O)_6]^{2+} + EDTA^{4-} \rightarrow [Cu(EDTA)]^{2-} + 6H_2O$: 2 particles become 7.
- $[Co(NH_3)_6]^{2+} + 3en \rightarrow [Co(en)_3]^{2+} + 6NH_3$: 4 particles become 7.
- More particles means more ways to arrange the energy, so **ΔS_{system} is positive**. The same number and type of dative bonds are broken and made (M–O or M–N), so ΔH is close to zero and $\Delta S_{surroundings}$ is small. ΔS_{total} is therefore positive and the chelate complex is favoured.
- **Uses of EDTA**: it binds metal ions so tightly that it is used to remove toxic heavy-metal ions from rivers and from the body, and to soften water in shampoos by trapping Ca^{2+}.

Working out formulae with silver nitrate

Chloride can be a ligand bonded to the metal or a free ion outside the complex. Silver nitrate precipitates only the **free** chloride ions as AgCl, so the mole ratio shows how many are outside. The three isomers of formula $CrCl_3 \cdot 6H_2O$ behave differently:

Complex	Mol AgCl per mol	Colour
$[Cr(H_2O)_6]Cl_3$	3	violet
$[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$	2	pale green
$[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$	1	dark green

Transition metals as catalysts

A catalyst speeds up a reaction by providing an alternative route with a lower activation energy, and is unchanged at the end. Transition metals are good catalysts because they can **change oxidation state** easily, and because they can use their 3d and 4s electrons to form weak bonds to reactants on their surface.

Heterogeneous catalysis

A **heterogeneous** catalyst is in a different phase from the reactants, usually a solid with gases or liquids. The reaction happens on its surface:

1. Reactant molecules are **adsorbed** onto active sites on the surface, forming weak bonds to the metal.
2. Bonds within the reactants are weakened, and the molecules are held close together in a favourable orientation, so they react more easily.
3. The products are **desorbed** from the surface, freeing the sites.

- **Strength of adsorption** must be just right: tungsten adsorbs too strongly, so products are not released; silver too weakly, so too few reactant molecules stick. Nickel and platinum are in between and make good catalysts.
- **Surface area**: the catalyst is spread thinly on a support, such as a ceramic honeycomb, to give a large surface area from a small amount of expensive metal.
- **Advantages**: solid catalysts are easy to separate from the products and suit continuous processes.
- **Examples**: iron in the Haber process; vanadium(V) oxide in the Contact process, which changes oxidation state and changes back: $SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$, then $2V_2O_4 + O_2 \rightarrow 2V_2O_5$; platinum, palladium and rhodium in catalytic converters, which remove carbon monoxide, nitrogen monoxide and unburnt hydrocarbons, for example $2CO + 2NO \rightarrow 2CO_2 + N_2$.
- **Developing better catalysts**: ethanoic acid is now made from methanol and carbon monoxide, $CH_3OH + CO \rightarrow CH_3COOH$, with 100% atom economy, using an iridium–iodide catalyst. Successive catalysts have allowed milder conditions and better selectivity; such advances are published in peer-reviewed journals so other scientists can check them.

Homogeneous catalysis

A **homogeneous** catalyst is in the same phase as the reactants. It works by forming an **intermediate**, usually by changing oxidation state and then changing back.

Iron ions catalysing the reaction between peroxodisulfate and iodide

1. Overall: $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$. Uncatalysed it is slow, because two negative ions repel each other, so collisions with enough energy are rare.
2. Step 1: $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$
3. Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$. The iron is back as Fe^{2+}, and Fe^{3+} works equally well because the steps can happen in either order.
4. Both steps are between oppositely charged ions, so they are faster. This works because $E^{\ominus}(Fe^{3+}/Fe^{2+}) = +0.77$ V lies **between** $E^{\ominus}(I_2/I^-) = +0.54$ V and $E^{\ominus}(S_2O_8^{2-}/SO_4^{2-}) = +2.01$ V, so both steps are feasible.

Electrode potentials show that a catalyst could work, but not that it will speed the reaction up: they say nothing about rates.

Autocatalysis

In **autocatalysis** a product catalyses the reaction. In the titration of ethanedioate with manganate(VII), $2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$, the reaction starts slowly (again two negative ions must collide), then speeds up as Mn^{2+} builds up, because Mn^{2+} provides a faster route:

- $4Mn^{2+} + MnO_4^- + 8H^+ \rightarrow 5Mn^{3+} + 4H_2O$
- $2Mn^{3+} + C_2O_4^{2-} \rightarrow 2Mn^{2+} + 2CO_2$
- The rate can be followed with a colorimeter measuring the purple colour, which does not disturb the reaction, or by titrating samples.

Quick checks

QUICK CHECK

What is the electron configuration of Fe^{3+}?

- [Ar] $3d^3\ 4s^2$
- [Ar] $3d^5$ ✓
- [Ar] $3d^6$
- [Ar] $3d^4\ 4s^1$

Iron is [Ar] $3d^6\ 4s^2$; the two 4s electrons are lost first, then one 3d electron.

QUICK CHECK

What is the coordination number of chromium in $[Cr(C_2O_4)_3]^{3-}$?

- 3
- 4
- 6 ✓
- 9

Each ethanedioate is bidentate, forming two dative bonds: 3 × 2 = 6. The coordination number counts bonds, not ligands.

QUICK CHECK

Why is a solution of $[Zn(H_2O)_6]^{2+}$ colourless?

- zinc has no d electrons
- its 3d sub-shell is full, so no d–d electron transition is possible
- water is a weak ligand
- the complex is tetrahedral

Zn^{2+} is $3d^{10}$. With every d orbital full, an electron cannot move to the upper set, so no visible light is absorbed.

QUICK CHECK

Excess aqueous ammonia is added to aqueous copper(II) sulfate. What is seen?

- a blue precipitate that stays
- a white precipitate
- a blue precipitate that dissolves to give a deep blue solution
- a yellow-green solution

A little ammonia deprotonates the aqua ion, giving blue $[Cu(OH)_2(H_2O)_4]$; excess ammonia then exchanges ligands to form deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$.

QUICK CHECK

Why are solutions of $[Fe(H_2O)_6]^{3+}$ more acidic than those of $[Fe(H_2O)_6]^{2+}$?

- Fe^{3+} has more water ligands
- Fe^{3+} has a higher charge density, so it weakens the O–H bonds in its water ligands more
- Fe^{3+} is a Brønsted–Lowry base
- Fe^{2+} forms a precipitate with water

Both have six water ligands. The smaller, more highly charged Fe^{3+} ion polarises them more, so they release H^+ more readily.

QUICK CHECK

Which reaction is favoured mainly by a large increase in entropy?

- $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CuCl_4]^{2-} + 6H_2O$
- $[Co(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 6H_2O$
- $[Ni(NH_3)_6]^{2+} + 3en \rightarrow [Ni(en)_3]^{2+} + 6NH_3$
- $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

Four particles become seven when bidentate en replaces ammonia: the chelate effect. ΔH is close to zero because Ni–N bonds are broken and made.

QUICK CHECK

Zinc and acid reduce vanadium(V) to vanadium(II), but tin only reduces it to vanadium(III). Why?

- tin is a stronger reducing agent than zinc
- V^{3+} is unstable
- $E^{\ominus}(Sn^{2+}/Sn)$ is more negative than $E^{\ominus}(VO^{2+}/V^{3+})$ but not more negative than $E^{\ominus}(V^{3+}/V^{2+})$
- tin forms a complex with V^{3+}

Sn^{2+}/Sn is −0.14 V. It can reduce couples with more positive $E^{\ominus}$ (+1.00 and +0.34 V) but not V^{3+}/V^{2+} at −0.26 V. Zinc, at −0.76 V, can reduce all three.

QUICK CHECK

In heterogeneous catalysis, why is silver a poor catalyst for many reactions?

- it adsorbs reactants too strongly
- it adsorbs reactants too weakly, so few stay on the surface long enough to react
- it is in the same phase as the reactants
- it changes oxidation state too easily

Too weak adsorption (silver) and too strong adsorption (tungsten) both make poor catalysts. Nickel and platinum adsorb with intermediate strength.

A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Arenes

Benzene has six carbons and three apparent double bonds, yet it refuses to react like an alkene. This topic explains why: its electrons are spread around the whole ring, which makes it unusually stable and steers it towards substitution rather than addition. It also covers the reactions that put new groups onto the ring, and why phenol reacts so much more readily.

The structure of benzene

Organic compounds are either **aliphatic** (straight or branched chains, and rings that are not aromatic) or **aromatic**, containing one or more benzene rings. Compounds containing a benzene ring are called **arenes**.

Benzene, C_6H_6, is a flat hexagonal ring. Each carbon forms σ bonds to two other carbons and one hydrogen, with bond angles of 120°. That leaves each carbon with one electron in a p orbital at right angles to the ring. The six p orbitals overlap sideways all the way round, forming rings of π electron density above and below the plane: the six π electrons are **delocalised** over all six carbons rather than held in three separate double bonds. The ring is drawn as a hexagon with a circle inside.

Evidence against Kekulé's structure

In 1865 Kekulé proposed a ring of alternating single and double bonds, cyclohexa-1,3,5-triene. Three pieces of evidence show that this is wrong:

- **Bond lengths**: X-ray diffraction shows all six C–C bonds are the same length, 0.139 nm, between a C–C single bond (0.154 nm) and a C=C double bond (0.134 nm). Kekulé's structure would have three of each.
- **Enthalpy of hydrogenation**: hydrogenating cyclohexene, with one C=C, releases 120 kJ mol^{-1}, so three C=C bonds should release 3 × 120 = 360 kJ mol^{-1}. Benzene actually releases only 208 kJ mol^{-1}. Benzene is **152 kJ mol^{-1} more stable** than Kekulé's structure: this is the **delocalisation energy**.
- **Reactivity**: alkenes decolourise bromine water quickly by addition, but benzene does not react with bromine water. Addition would break up the stable delocalised system.

Naming aromatic compounds

- Simple derivatives are named with benzene as the parent: methylbenzene, chlorobenzene, nitrobenzene, benzenecarboxylic acid (C_6H_5COOH), benzaldehyde (C_6H_5CHO).
- With two or more substituents, number the ring to give the lowest numbers, and list different groups alphabetically: 1,3-dimethylbenzene, 1-chloro-4-methylbenzene, 2-methyl-1,3,5-trinitrobenzene (TNT).
- When the ring is attached to a more important group, it is called the **phenyl** group, C_6H_5–: phenylamine ($C_6H_5NH_2$), phenylethene, phenylethanone ($C_6H_5COCH_3$), phenyl ethanoate.

- **Toxicity**: benzene is a carcinogen, so methylbenzene is used instead in school laboratories.

Combustion and hydrogenation

- **Combustion**: $C_6H_6 + 7½O_2 \rightarrow 6CO_2 + 3H_2O$. Arenes burn with a very **smoky, sooty flame** because their high carbon-to-hydrogen ratio leaves unburnt carbon.
- **Hydrogenation**: with hydrogen over a nickel catalyst at about 200 °C and 30 atm, benzene adds three molecules of H_2 to give cyclohexane. This is one of the few addition reactions of benzene, and it needs far harsher conditions than for an alkene.

Electrophilic substitution

The delocalised π electrons give benzene a region of high electron density that attracts **electrophiles**. But addition would destroy the delocalisation, so instead benzene undergoes **electrophilic substitution**: one hydrogen is replaced and the stable ring is restored. Every mechanism follows the same three stages:

1. **Making the electrophile**: a catalyst turns the reagent into a positive ion, E^+.
2. **Attack**: a pair of π electrons from the ring forms a bond to E^+. This gives a positively charged intermediate in which the delocalisation covers only five carbons (drawn as a horseshoe from carbon 2 to carbon 6), with the carbon that was attacked holding both H and E.
3. **Loss of H^+**: the electrons of the C–H bond return to the ring, restoring the full delocalised system, and H^+ is released, usually regenerating the catalyst.

Reaction	Reagents and conditions	Electrophile and how it forms	Product
bromination (halogenation)	bromine with an iron(III) bromide catalyst (or iron filings, which form $FeBr_3$); room temperature. Chlorine with $AlCl_3$ or $FeCl_3$ works the same way.	Br^+: $Br_2 + FeBr_3 \rightarrow Br^+ + FeBr_4^-$; then $H^+ + FeBr_4^- \rightarrow FeBr_3 + HBr$	bromobenzene + HBr
nitration	concentrated nitric acid and concentrated sulfuric acid, below about 55 °C (higher temperatures add more nitro groups)	NO_2^+, the nitronium ion: $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$; then $H^+ + HSO_4^- \rightarrow H_2SO_4$	nitrobenzene
sulfonation	fuming sulfuric acid (SO_3 in concentrated H_2SO_4), heated under reflux	SO_3, whose sulfur carries a large δ+ charge from its three oxygens	benzenesulfonic acid, $C_6H_5SO_3H$ (used for detergents, dyes and drugs)

Friedel–Crafts alkylation	a halogenoalkane with an anhydrous aluminium chloride catalyst, heated under reflux	R^+: $CH_3CH_2Cl + AlCl_3 \rightarrow CH_3CH_2^+ + AlCl_4^-$; then $H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$	ethylbenzene + HCl
Friedel–Crafts acylation	an acyl chloride with an anhydrous $AlCl_3$ catalyst, heated under reflux at about 50 °C	RCO^+, the acylium ion: $CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$	phenylethanone, $C_6H_5COCH_3$, + HCl

EXAM TIP

Friedel–Crafts reactions matter in synthesis because they form a new **carbon–carbon bond** to the ring. Nitration is the first step towards phenylamine, used to make dyes, and towards explosives such as TNT.

COMMON MISTAKE

The aluminium chloride must be **anhydrous**: water would react with it and destroy the catalyst. In the mechanism, the curly arrow for the attack starts from the **circle** (the delocalised electrons), not from a single bond.

How side groups change the ring

Electron-releasing groups such as alkyl groups, –OH and $–NH_2$ push electron density into the ring, making it more attractive to electrophiles. Methylbenzene therefore nitrates more readily than benzene, under milder conditions.

When an atom with a lone pair (O, N or Cl) is bonded directly to the ring, its lone pair overlaps with the π system and becomes partly delocalised into it. This changes both the ring and the side group:

- **Chlorobenzene**: the C–Cl bond gains some double-bond character and becomes stronger, so chlorobenzene does not undergo the substitution and elimination reactions of halogenoalkanes; the electron-rich ring also repels nucleophiles.
- **Phenol**: the C–O bond becomes stronger and the O–H bond weaker, so phenol is more acidic than an alcohol and is not oxidised like one.
- **Phenylamine**: the nitrogen lone pair is partly delocalised, so it is less available to accept a proton, and phenylamine is a much weaker base than aliphatic amines (see Topic 19).

Phenol

In **phenol**, C_6H_5OH, the –OH group is bonded directly to the ring. ($C_6H_5CH_2OH$, with a CH_2 in between, is an ordinary alcohol, phenylmethanol.)

Phenol as an acid

The phenoxide ion is stabilised because its negative charge is partly delocalised into the ring, so phenol is a **weak acid**: stronger than alcohols and water, but weaker than carboxylic acids.

Reagent	Ethanol	Phenol	Ethanoic acid
sodium	reacts: H_2	reacts: $C_6H_5OH + Na \rightarrow C_6H_5O^-Na^+ + ½H_2$	reacts: H_2
sodium hydroxide	no reaction	reacts: $C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O$ (the solid dissolves, as sodium phenoxide is soluble)	reacts
sodium carbonate	no reaction	**no reaction**: phenol is too weak an acid	fizzes: CO_2

The sodium carbonate test therefore tells a phenol from a carboxylic acid.

Phenol with bromine water

Phenol decolourises bromine water at room temperature, with no catalyst, giving a **white precipitate** of 2,4,6-tribromophenol:

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$$

The oxygen lone pair delocalised into the ring raises its electron density so much that it **polarises** bromine molecules directly, without needing a halogen carrier, and three hydrogens are substituted rather than one. Benzene needs $FeBr_3$ and pure bromine, and substitutes only once. Phenols are used to make plastics, resins, antiseptics and disinfectants.

Quick checks

QUICK CHECK

Which observation is the best evidence that benzene's six C–C bonds are identical?

- benzene burns with a smoky flame
- X-ray diffraction shows all six C–C bonds are 0.139 nm long
- benzene is a liquid at room temperature
- benzene reacts with hydrogen over nickel

Kekulé's structure would have three short C=C and three long C–C bonds. One intermediate length shows the π electrons are spread evenly round the ring.

QUICK CHECK

The enthalpy of hydrogenation of cyclohexene is −120 kJ mol^{-1} and of benzene −208 kJ mol^{-1}. What is the delocalisation energy of benzene?

- 88 kJ mol^{-1}
- 208 kJ mol^{-1}
- 152 kJ mol^{-1}
- 328 kJ mol^{-1}

Three C=C bonds would release 3 × 120 = 360 kJ mol^{-1}; benzene releases 208, so it is 360 − 208 = 152 kJ mol^{-1} more stable. 88 is 208 − 120, which compares with only one double bond.

QUICK CHECK

Why does benzene undergo substitution rather than addition with electrophiles?

- benzene has no double bonds at all
- electrophiles cannot attack benzene
- substitution keeps the stable delocalised π system, which addition would destroy
- addition products are too unstable to isolate

Losing H^+ at the end restores the full ring of delocalised electrons and its delocalisation energy.

QUICK CHECK

What is the electrophile in the nitration of benzene, and how is it formed?

- NO_3^-, from nitric acid losing H^+
- NO_2^+, from $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$
- H^+, from sulfuric acid
- NO_2, a radical made by heating

Sulfuric acid, the stronger acid, protonates nitric acid, which then loses water to give the nitronium ion. Sulfuric acid is regenerated at the end.

QUICK CHECK

Which reagents would convert benzene into phenylethanone, $C_6H_5COCH_3$?

- ethanol and concentrated sulfuric acid
- chloroethane and $AlCl_3$
- ethanoyl chloride and anhydrous $AlCl_3$
- ethanoic acid and $FeBr_3$

Friedel–Crafts acylation: $AlCl_3$ forms the acylium ion CH_3CO^+ from ethanoyl chloride. Chloroethane would give ethylbenzene by alkylation.

QUICK CHECK

How can you distinguish phenol from ethanoic acid?

- add sodium: only phenol fizzes
- add sodium hydroxide: only phenol reacts
- add sodium carbonate: only ethanoic acid fizzes
- measure the pH: both are neutral

Both react with sodium and with sodium hydroxide, but phenol is too weak an acid to release CO_2 from a carbonate.

QUICK CHECK

Why does phenol react with bromine water without a catalyst, while benzene does not?

- phenol is a stronger acid than benzene
- a lone pair on oxygen is delocalised into the ring, raising its electron density enough to polarise Br_2
- the OH group is an electrophile
- phenol has a double bond outside the ring

The extra electron density makes the ring so attractive to electrophiles that Br_2 is polarised directly, and substitution happens at three positions to give 2,4,6-tribromophenol.

QUICK CHECK

In the mechanism of electrophilic substitution, what does the "horseshoe" in the intermediate show?

- the electrophile is attached to every carbon
- the remaining delocalised electrons and the positive charge are spread over five carbons
- the ring has opened
- the hydrogen has already been lost

Two of the six π electrons have formed the bond to the electrophile, so the other four, with the positive charge, are delocalised over the remaining five carbons. The sixth carbon carries both H and E.

A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Organic nitrogen compounds

Nitrogen's lone pair is the key to this topic. It makes amines basic, lets them act as ligands and nucleophiles, and joins monomers into nylon, Kevlar and proteins. You will also meet azo dyes, the zwitterions of amino acids, and the difference between addition and condensation polymers.

Naming amines and amides

- **Primary amines** have the –NH_2 group. $CH_3CH_2CH_2NH_2$ is called propylamine (the name the exam board often uses) or propan-1-amine (the IUPAC name, built like an alcohol name). With a higher-priority group present, the prefix **amino-** is used: $CH_3CH(NH_2)COOH$ is 2-aminopropanoic acid.
- **Secondary and tertiary amines**: the longest chain gives the name and other alkyl groups on nitrogen are given the locant N: $CH_3CH_2CH_2NHCH_3$ is N-methylpropylamine (N-methylpropan-1-amine); $(CH_3)_2NCH_2CH_2CH_3$ is N,N-dimethylpropylamine. If both groups are the same, $(CH_3CH_2)_2NH$ is diethylamine.
- **Diamines**: $H_2N(CH_2)_6NH_2$ is hexane-1,6-diamine.
- **Amides** end in -amide: CH_3CONH_2 is ethanamide; $CH_3CH_2CONHCH_3$ is N-methylpropanamide; $CH_3CH_2CON(CH_3)_2$ is N,N-dimethylpropanamide.

Amines as bases

Small amines smell fishy and dissolve readily in water, because the N–H groups and the nitrogen lone pair hydrogen bond with water molecules. The lone pair on nitrogen can accept a proton, so amines are **Brønsted–Lowry bases**:

$$CH_3NH_2(aq) + H_2O(l) \rightleftharpoons CH_3NH_3^+(aq) + OH^-(aq)$$

How strong a base an amine is depends on how available its lone pair is:

- **Aliphatic amines are stronger bases than ammonia**: alkyl groups are electron-releasing, pushing electron density onto the nitrogen and making its lone pair more available. Secondary amines, with two alkyl groups, are stronger still: diethylamine > ethylamine > ammonia.
- **Phenylamine is a much weaker base than ammonia**: its lone pair is partly delocalised into the benzene ring, so it is less available to accept a proton. Its solutions are barely alkaline.
- **Amides are not basic**: the nitrogen lone pair is delocalised towards the electronegative C=O group.

Salts, buffers and complexes

- **With acids** amines form ammonium salts: $CH_3NH_2 + HCl \rightarrow CH_3NH_3^+Cl^-$ (methylammonium chloride); $2CH_3NH_2 + H_2SO_4 \rightarrow (CH_3NH_3^+)_2SO_4^{2-}$. These ionic salts are crystalline solids that dissolve in water even

when the amine does not: phenylamine is only slightly soluble, but phenylammonium chloride is soluble. Adding NaOH turns the salt back into the amine.

- **Basic buffers** can be made from an amine and its salt, such as methylamine with methylammonium chloride, just as with ammonia and ammonium chloride.
- **As ligands**, amines form complexes with transition metal ions, like ammonia does: $[Cu(H_2O)_6]^{2+} + 4CH_3CH_2NH_2 \rightarrow [Cu(CH_3CH_2NH_2)_4(H_2O)_2]^{2+} + 4H_2O$, a deep blue solution.

Amines as nucleophiles

With halogenoalkanes

The lone pair also makes amines nucleophiles. A primary amine attacks a halogenoalkane in nucleophilic substitution to give a secondary amine; a second amine molecule removes a proton from the intermediate:

$$2CH_3CH_2NH_2 + CH_3CH_2Br \rightarrow (CH_3CH_2)_2NH + CH_3CH_2NH_3^+Br^-$$

The product still has a lone pair, so it can react again, giving a tertiary amine and finally a **quaternary ammonium salt**, $R_4N^+X^-$, in which the nitrogen has no lone pair left. Mixtures of products are normal.

With acyl chlorides

Amines react with acyl chlorides at room temperature to give N-substituted amides (an addition–elimination reaction, see Topic 15): $CH_3COCl + 2CH_3NH_2 \rightarrow CH_3CONHCH_3 + CH_3NH_3^+Cl^-$. Phenylamines react in the same way; paracetamol is made by acylating the $-NH_2$ group of 4-aminophenol with an ethanoyl group.

Making amines

Method	Reagents and conditions	Comment
halogenoalkane + ammonia	excess ammonia dissolved in ethanol, heated in a sealed tube: $CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$	one step, but the amine formed reacts further, giving a mixture of amines. A large excess of ammonia makes the primary amine the main product.
via a nitrile	1. KCN in aqueous ethanol, heat under reflux: $CH_3CH_2Br + CN^- \rightarrow CH_3CH_2CN + Br^-$. 2. $LiAlH_4$ in dry ether (or H_2 with a nickel catalyst): $CH_3CH_2CN + 4[H] \rightarrow CH_3CH_2CH_2NH_2$	gives only the primary amine, with one extra carbon; but it is two steps (lower overall yield) and KCN is highly toxic

reducing a nitroarene	tin (or iron) and concentrated hydrochloric acid, heat: $C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$	in the acid the product is the salt $C_6H_5NH_3^+Cl^-$; add NaOH to release phenylamine, then separate it by steam distillation

Azo dyes

Making an azo dye from phenylamine

1. **Diazotisation**: react phenylamine with nitrous acid below 10 °C. Nitrous acid is unstable, so it is made in the flask from sodium nitrite and hydrochloric acid: $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$.

2. $C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O$, forming benzenediazonium chloride, which contains the $–N{\equiv}N^+$ group.

   Keep it cold: above about 10 °C the diazonium ion reacts with water, $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2 + H^+$, releasing nitrogen gas. Much below 0 °C the reaction is too slow.

3. **Coupling**: add the diazonium salt to phenol dissolved in sodium hydroxide. The diazonium ion is a weak electrophile that attacks the activated ring: $C_6H_5N_2^+ + C_6H_5OH \rightarrow C_6H_5–N{=}N–C_6H_4OH + H^+$.

4. The product, 4-hydroxyazobenzene, is a yellow-orange **azo dye**. The –N=N– link joins two benzene rings into one long delocalised system that absorbs visible light. Azo compounds are used as dyes, pigments and food colourings.

Polymers: addition and condensation

Addition polymers

Monomers with a C=C double bond join without losing any atoms. Draw the monomer with its other groups arranged around the double bond, then open the double bond to form the repeat unit. Propenamide, $CH_2{=}CHCONH_2$, gives poly(propenamide), $–[CH_2–CH(CONH_2)]–$. Poly(ethenol), $–[CH_2–CH(OH)]–$, has an –OH group on every other carbon, so it forms many hydrogen bonds with water and dissolves: it is used for soluble laundry bags and detergent capsules. Most poly(alkenes), with only strong non-polar C–C and C–H bonds, are unreactive and not biodegradable.

Condensation polymers: polyamides

In condensation polymerisation, two different functional groups react and a small molecule (water or HCl) is lost at each link. A **polyamide** forms from a dicarboxylic acid (or diacyl dichloride) and a diamine, linked by **amide groups**, –CONH–:

- **Nylon 6,6**: hexanedioic acid + hexane-1,6-diamine. Repeat unit $–CO(CH_2)_4CONH(CH_2)_6NH–$. The "6,6" counts the six carbons in each monomer.
- **Kevlar**: benzene-1,4-dicarboxylic acid + benzene-1,4-diamine. Repeat unit $–CO–C_6H_4–CONH–C_6H_4–NH–$. Its flat chains line up and hydrogen bond strongly, making it extremely strong: it is used in body armour.
- One monomer with an amine group at one end and an acid group at the other can also polymerise on its own, as amino acids do to make proteins.
- Asked for the *type of polymer*, say polyamide or polyester; asked for the *type of polymerisation*, say condensation.

- **Intermolecular forces**: polyamide chains hydrogen bond to each other between the N–H of one chain and the C=O of another, as well as having dipole–dipole and London forces. Polyesters have no N–H, so only dipole–dipole and London forces act between chains; polyamides therefore have higher melting points.
- **Hydrolysis**: the polar amide (and ester) links can be attacked, so condensation polymers are biodegradable. Acid hydrolysis of nylon 6,6 gives hexanedioic acid and the diammonium salt of the diamine ($H_3N^+(CH_2)_6NH_3^+$); alkaline hydrolysis gives the diamine and the salt of the acid ($^-OOC(CH_2)_4COO^-$).

Amino acids

An α-amino acid has an amine group and a carboxylic acid group on the same carbon, $H_2NCH(R)COOH$. Glycine, the simplest, has R = H: aminoethanoic acid. Others are named systematically, for example 2-amino-3-hydroxypropanoic acid (R = CH_2OH) or 2-aminobutanedioic acid (aspartic acid, with a second –COOH in the R group, an acidic amino acid). Lysine has a second $–NH_2$ and is a basic amino acid.

- **Optical activity**: every amino acid except glycine has four different groups on the central carbon, so it is chiral and its enantiomers rotate plane-polarised light in opposite directions. Glycine has two hydrogens on that carbon.
- **Zwitterions**: in the solid and in neutral solution, the acid group donates its proton to the amine group, giving a **zwitterion**, $^+H_3NCH(R)COO^-$, with a positive and a negative charge but no overall charge. Strong ionic attractions between zwitterions give amino acids their high melting points, far above those of molecules held only by hydrogen bonds.
- **Acid–base behaviour**: in acid the COO^- group gains a proton, giving the cation $^+H_3NCH(R)COOH$; in alkali the NH_3^+ group loses one, giving the anion $H_2NCH(R)COO^-$. With glycine: $^+H_3NCH_2COO^- + HCl \rightarrow {}^+H_3NCH_2COOH\ Cl^-$, and $^+H_3NCH_2COO^- + NaOH \rightarrow H_2NCH_2COO^-Na^+ + H_2O$. Because they can take up both, amino acid solutions act as weak buffers. Extra acid or amine groups in the R group change state too.
- Their functional groups also react normally: for example, heating with methanol and an acid catalyst esterifies the –COOH group.

Peptides and proteins

- The –COOH of one amino acid and the $–NH_2$ of another condense, losing water, to form a **peptide link**, –CONH– (an amide group). Two amino acids give a **dipeptide**; because either one can provide the –COOH, two different amino acids can form **two** different dipeptides.
- **Proteins** are long polypeptide chains, condensation polymers of amino acids.
- The chains fold into spirals (α-helices) and sheets held in shape by **hydrogen bonds** between the N–H of one peptide link and the C=O of another. This three-dimensional shape is vital to a protein's job, such as an enzyme's active site.
- Heating a protein with 6 mol dm^{-3} hydrochloric acid under reflux hydrolyses the peptide links, releasing the amino acids, which can then be identified by chromatography with ninhydrin (Topic 15).

Quick checks

QUICK CHECK

Which is the strongest base?

- ammonia
- phenylamine
- diethylamine
- ethanamide

Two electron-releasing ethyl groups make diethylamine's lone pair the most available. Phenylamine's lone pair is delocalised into the ring and ethanamide's towards the C=O, so both are much weaker.

QUICK CHECK

Why is a large excess of ammonia used to make ethylamine from bromoethane?

- ammonia is a catalyst
- so that bromoethane is more likely to meet ammonia than the ethylamine already formed, reducing further substitution
- to neutralise the HBr formed
- ammonia is the solvent

Ethylamine is itself a nucleophile and can react with more bromoethane to give secondary and tertiary amines. Swamping it with ammonia makes the primary amine the main product.

QUICK CHECK

Why must the diazotisation of phenylamine be kept below 10 °C?

- nitrous acid boils at 10 °C
- the azo dye decomposes
- above 10 °C the diazonium ion reacts with water to form phenol and nitrogen gas
- phenylamine freezes at higher temperatures

Benzenediazonium ions are unstable when warm. The low temperature keeps them long enough to couple with phenol.

QUICK CHECK

How many molecules of water are released when 50 molecules of hexanedioic acid react with 50 molecules of hexane-1,6-diamine to form one nylon 6,6 chain?

- 50
- 99
- 100
- 101

100 monomers joined in a single chain need 99 amide links, each releasing one water molecule. (For large n, 2n − 1 is usually written as 2n.)

QUICK CHECK

Why do amino acids have high melting points?

- they are large molecules with strong London forces
- they form covalent networks
- they exist as zwitterions, held together by strong ionic attractions
- they hydrogen bond more than water

In the solid, each molecule carries NH_3^+ and COO^- groups, so the crystal is held together like an ionic lattice.

QUICK CHECK

Which form of glycine is present at pH 1?

- $H_2NCH_2COO^-$
- H_2NCH_2COOH
- $^+H_3NCH_2COOH$
- $^+H_3NCH_2COO^-$

In strong acid both groups are protonated, giving the cation. H_2NCH_2COOH, with no charges, is never the main form in solution.

QUICK CHECK

Why do polyamides usually have higher melting points than polyesters of similar chain length?

- polyamides have stronger covalent bonds in the chain
- polyamide chains hydrogen bond to each other through N–H and C=O groups
- polyesters have no permanent dipoles
- polyamides are addition polymers

Both have C=O dipoles, but only polyamides have N–H groups able to hydrogen bond to C=O groups in neighbouring chains.

QUICK CHECK

Glycine (G) and alanine (A) are joined by a single peptide link. How many different dipeptides can form from one of each?

1

2

3

4

Gly–Ala and Ala–Gly are different: either amino acid can provide the –COOH that forms the peptide link. (Allowing Gly–Gly and Ala–Ala as well would give four.)

A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Organic synthesis

This topic pulls the organic chemistry of the course together. It shows how to plan a route from one compound to another, how to lengthen a carbon chain with Grignard reagents, and the practical techniques used to make, separate, purify and identify an organic product safely.

Planning a synthetic route

To plan a synthesis, identify the functional group you start with and the one you need, then work out a sequence of known reactions that connects them, often working backwards from the target. For each step you must be able to give the **reagents**, the **conditions** and the **type of reaction**. Fewer steps usually means a higher overall yield, because some product is lost at every stage.

Key aliphatic conversions

From → to	Reagents and conditions	Type of reaction
alkane → halogenoalkane	Cl_2 or Br_2, UV light	free-radical substitution
alkene → alkane	H_2, nickel catalyst, heat	addition (reduction)
alkene → halogenoalkane / dihalogenoalkane	HBr or Br_2, room temperature	electrophilic addition
alkene → alcohol	steam, phosphoric(V) acid catalyst, high temperature and pressure	hydration (addition)
alkene → diol	acidified potassium manganate(VII)	oxidation
halogenoalkane → alcohol	aqueous NaOH or KOH, heat under reflux	nucleophilic substitution
halogenoalkane → alkene	KOH in ethanol, heat under reflux	elimination
halogenoalkane → nitrile	KCN in aqueous ethanol, heat under reflux	nucleophilic substitution (adds one carbon)
halogenoalkane → amine	excess ammonia in ethanol, heat in a sealed tube	nucleophilic substitution

alcohol → halogenoalkane	PCl_5; or KBr with 50% H_2SO_4; or red phosphorus and iodine	substitution
alcohol → alkene	concentrated H_3PO_4 (or H_2SO_4), heat	elimination (dehydration)
primary alcohol → aldehyde	$K_2Cr_2O_7$ and dilute H_2SO_4, heat gently and distil off the aldehyde as it forms	oxidation
primary alcohol or aldehyde → carboxylic acid	excess $K_2Cr_2O_7$ and dilute H_2SO_4, heat under reflux	oxidation
secondary alcohol → ketone	$K_2Cr_2O_7$ and dilute H_2SO_4, heat under reflux	oxidation
aldehyde, ketone or carboxylic acid → alcohol	$LiAlH_4$ in dry ether	reduction
aldehyde or ketone → hydroxynitrile	HCN with KCN	nucleophilic addition (adds one carbon)
nitrile → carboxylic acid	dilute HCl, heat under reflux	hydrolysis
nitrile → amine	$LiAlH_4$ in dry ether, or H_2 with a nickel catalyst	reduction
carboxylic acid → acyl chloride	PCl_5, room temperature	substitution
carboxylic acid → ester	alcohol with concentrated H_2SO_4, heat	esterification (condensation)
acyl chloride → acid / ester / amide / N-substituted amide	water / alcohol / concentrated ammonia / primary amine, all at room temperature	addition–elimination
ester → carboxylic acid (or its salt) + alcohol	dilute acid or aqueous NaOH, heat under reflux	hydrolysis

Aromatic routes

- **Benzene** → **nitrobenzene** → **phenylamine**: concentrated HNO_3 and H_2SO_4 below 55 °C; then tin and concentrated HCl, heat, followed by NaOH.
- **Phenylamine** → **azo dye**: $NaNO_2$ and HCl below 10 °C, then phenol in NaOH. Or phenylamine + an acyl chloride gives an N-phenyl amide.
- **Benzene** → **phenylethanone** (CH_3COCl with $AlCl_3$), which then reacts like any ketone: $LiAlH_4$ gives 1-phenylethanol, which can be esterified; HCN/KCN gives a hydroxynitrile, which hot HCl hydrolyses to a hydroxy acid.

- **Other ring reactions**: $Br_2/FeBr_3$, a chloroalkane with $AlCl_3$, SO_3 in fuming sulfuric acid, and H_2 over nickel (see Topic 18).

Grignard reagents

A **Grignard reagent**, RMgX, is made by warming a halogenoalkane with magnesium turnings in **dry ether**: $CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr$ (ethylmagnesium bromide). The carbon bonded to magnesium carries a partial negative charge, so it acts as a **nucleophilic carbon** and forms new carbon–carbon bonds, lengthening the chain. Everything must be dry, because water destroys the reagent.

The Grignard reagent adds to the δ+ carbon of a C=O group; adding water (or dilute acid) afterwards gives the product.

Grignard reagent reacts with	Product (after adding water)	Example with CH_3CH_2MgBr
methanal, HCHO	primary alcohol	$CH_3CH_2CH_2OH$, propan-1-ol
another aldehyde, R′CHO	secondary alcohol	with ethanal: $CH_3CH_2CH(OH)CH_3$, butan-2-ol
a ketone, R′COR″	tertiary alcohol	with propanone: $CH_3CH_2C(OH)(CH_3)_2$, 2-methylbutan-2-ol
carbon dioxide	carboxylic acid	CH_3CH_2COOH, propanoic acid

The general equations, with the magnesium ending up as Mg(OH)X: RMgX + HCHO → RCH_2OH; RMgX + R′CHO → RCH(OH)R′; RMgX + R′COR″ → RR′C(OH)R″; RMgX + CO_2 → RCOOH (each after adding H_2O). Other ways to add a carbon are reacting a halogenoalkane with KCN, or a carbonyl with HCN.

Practical techniques

Heating under reflux and distillation

- **Reflux** heats a reaction mixture for a long time without losing volatile substances: a vertical condenser turns vapours back to liquid, which drips back into the flask. It is used, for example, to oxidise a primary alcohol fully to a carboxylic acid with excess dichromate. **Never seal the top** of the condenser: pressure would build up and the apparatus could explode.
- **Distillation** separates a liquid product from the mixture, collecting the fraction that boils near the product's boiling point; the thermometer bulb sits level with the side arm. Distilling the aldehyde off as it forms, with limited dichromate, stops it being oxidised further.
- **Anti-bumping granules** make small bubbles form, preventing violent uneven boiling. Organic liquids are heated with electric heaters or water baths, as they are usually flammable.

- When drawing apparatus: no lines across the joins between flask, still head and condenser; the condenser has an outer water jacket sealed at both ends, with water in at the bottom and out at the top.

Purifying an organic liquid

1. Put the impure product in a **separating funnel** and shake it with sodium hydrogencarbonate solution to neutralise acid impurities, releasing the pressure of CO_2 often. Shaking with saturated sodium chloride solution helps the layers separate.
2. Let the layers separate and run off the aqueous layer. The organic layer is usually the less dense, upper layer, but check densities.
3. Run the organic layer into a dry flask and add a **drying agent**, such as anhydrous calcium chloride or magnesium sulfate, until the liquid is clear. The drying agent must be insoluble in, and not react with, the product.
4. Decant or filter the liquid off the drying agent and **distil**, collecting the fraction at the product's boiling point.

Purifying an organic solid: recrystallisation

Step	Reason
Dissolve the impure solid in the **minimum volume of hot solvent**.	The solvent should dissolve the compound well when hot but poorly when cold. Using the minimum gives a saturated solution, so crystals form on cooling.
Filter the hot solution quickly through a fluted filter paper.	Removes insoluble impurities; keeping it hot stops crystals forming in the funnel.
Cool the filtrate, finally in ice.	The compound crystallises; soluble impurities stay dissolved because there is too little of them to saturate the solution.
Collect the crystals by suction filtration with a Büchner funnel.	Reduced pressure makes filtration fast.
Wash with a little cold solvent and dry between filter papers.	Removes soluble impurities on the surface without dissolving much product. Wet crystals give a mass that is too high, even a "yield" above 100%.

Steam distillation and solvent extraction

- **Steam distillation**: steam is blown through the mixture and the product distils over with the water at below 100 °C, so a high-boiling compound that would decompose at its own boiling point can be separated. Phenylamine is purified this way.
- **Solvent extraction**: shake the mixture with an organic solvent in which the product dissolves better than in water, separate the layers, dry the organic layer and distil off the solvent.

Checking purity

- **Melting point**: a pure solid melts sharply at the value in data books; impurities lower the melting point and make it melt over a range. Use a melting-point apparatus, or a capillary tube strapped to a thermometer in an oil bath (the oil must boil above the sample's melting point). Heat slowly so the thermometer and sample stay at the same temperature.
- **Boiling point**: measured with the thermometer bulb in the vapour just above the boiling liquid. It is less useful for identification, since different liquids can share a boiling point, and it changes with atmospheric pressure.

Hazards and risks

- A **hazard** is anything with the potential to cause harm: toxic, flammable, harmful, irritant, corrosive, oxidising, carcinogenic. A **risk** is the chance that harm actually results, which depends on the amount, the concentration and how the substance is used.
- Control measures match the hazard: eye protection for irritants and corrosives, gloves and a fume cupboard for toxic substances, no naked flames near flammable liquids, oxidising agents kept away from flammable materials.

Identifying the product

Combustion analysis

Empirical formula from combustion

1. 0.290 g of a compound of C, H and O burns completely to give 0.660 g of CO_2 and 0.270 g of H_2O. Find its empirical formula.
2. Carbon: moles of CO_2 = 0.660 ÷ 44.0 = 0.0150 mol, so 0.0150 mol of C, mass 0.180 g.
3. Hydrogen: moles of H_2O = 0.270 ÷ 18.0 = 0.0150 mol, so 2 × 0.0150 = 0.0300 mol of H, mass 0.030 g.
4. Oxygen by difference: 0.290 − 0.180 − 0.030 = 0.080 g, so 0.080 ÷ 16.0 = 0.0050 mol.
5. Ratio C : H : O = 0.0150 : 0.0300 : 0.0050 = 3 : 6 : 1, so the empirical formula is $\mathbf{C_3H_6O}$. If the mass spectrum shows M^+ at 58, this is also the molecular formula: propanone or propanal, which IR, NMR or chemical tests can tell apart.

The spectroscopic methods (mass spectra, IR and NMR) are covered in Topics 10 and 15, including a full worked identification.

Chemical tests for functional groups

Functional group	Test	Positive result
alkene	bromine water	orange to colourless
alcohol or carboxylic acid (–OH)	PCl_5	steamy fumes of HCl
alcohol, phenol or carboxylic acid	sodium metal	fizzing (hydrogen)
aldehyde or ketone	2,4-DNPH	orange precipitate
aldehyde	Fehling's or Benedict's solution, warm	blue to brick-red precipitate
aldehyde	Tollens' reagent, warm	silver mirror
$CH_3CO–$ or $CH_3CH(OH)–$	iodine and NaOH, warm	pale yellow precipitate of CHI_3
carboxylic acid	sodium carbonate or hydrogencarbonate	fizzing (CO_2)
primary or secondary alcohol, aldehyde	acidified $K_2Cr_2O_7$, warm	orange to green
halogenoalkane	warm with NaOH(aq), acidify with HNO_3, add $AgNO_3$	white (Cl), cream (Br) or yellow (I) precipitate, forming slowly
acyl chloride	water, then $AgNO_3$	steamy fumes, then an immediate white precipitate

Quick checks

QUICK CHECK

Which sequence converts bromoethane into propanoic acid?

- aqueous NaOH, then acidified dichromate
- KCN in ethanol, then heat with dilute HCl
- ammonia, then $LiAlH_4$
- KOH in ethanol, then steam

Cyanide adds a carbon (CH_3CH_2CN), and acid hydrolysis turns the nitrile into CH_3CH_2COOH. NaOH then oxidation would give ethanoic acid, with only two carbons.

QUICK CHECK

Which product forms when CH_3MgI reacts with propanone and the mixture is then treated with water?

- propan-2-ol
- butan-2-ol
- 2-methylpropan-2-ol
- butanoic acid

A Grignard reagent with a ketone gives a tertiary alcohol: the CH_3 joins the carbonyl carbon, giving $(CH_3)_3COH$.

QUICK CHECK

Why is a primary alcohol oxidised under reflux to make a carboxylic acid, but distilled to make an aldehyde?

- reflux removes water
- reflux keeps the aldehyde in the flask to be oxidised further; distilling removes it as soon as it forms
- distillation is faster
- the aldehyde needs a higher temperature

The aldehyde has a lower boiling point than the alcohol and acid, so distillation carries it away before it meets more oxidising agent.

QUICK CHECK

In recrystallisation, why is the minimum volume of hot solvent used?

- to save solvent
- to dissolve the impurities
- so the solution is saturated and the product crystallises on cooling
- so the solution cools faster

With too much solvent, much of the product stays dissolved when cold and the yield falls, or no crystals form at all.

QUICK CHECK

A student's crystals melt between 118 °C and 124 °C; the data book value is 127 °C. What does this suggest?

- the crystals are pure
- the crystals are impure: impurities lower the melting point and widen the range
- the thermometer reads too high
- the compound has decomposed into a purer form

A pure substance melts sharply at its data book value. A low, broad melting range points to impurities, such as remaining solvent.

QUICK CHECK

Which statement about hazard and risk is correct?

- a hazard is the chance of harm
- a dilute solution of a corrosive substance has the same risk as the concentrated one
- risk depends on how much of a hazardous substance is used and how
- flammable liquids are a risk only when heated with electric heaters

A hazard is the potential to cause harm; risk is the chance that harm actually happens, which falls with smaller amounts, lower concentrations and good control measures.